

Production planning and control

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What do you mean by production planning & control??

Production planning and control can be viewed as the nervous system of the production operation. This function aims at efficient utilization of material, resources, people and facilities in any undertaking through planning, coordination and controlling the production activities that transform the raw materials into finished products or components as a most optimal manner. All the activities in the manufacturing or production cycle must be planned, coordinated, organized and controlled to its objectives.

Definition

- According to Charles A. Koepke, “Production planning and control may be defined as the coordination of a series of functions according to a plan which will, economically utilize the plant facilities and regulate the orderly movement of goods through their entire manufacturing cycle, from the procurement of all materials to the shipping of finished goods at a predetermined rate”

objectives

1. Nature of the inputs

To manufacture a product, different types of inputs are used. The quality of the product depends upon the nature of the inputs used.

2. Quantity of inputs

A product can be prepared only when there is an estimate of the required composition of inputs.

3. Proper coordination

It ensures the proper coordination among the work force, machine and equipment. This leads to avoidance of wastages and smooth flow of production.

4. Better Control

Production planning is a method of control. For a better control, planning is a precondition. Only then one can compare the performance and calculate the deviations which lead control of the production.

Objectives continues...

5. Ensures uninterrupted production

The planning of materials ensures the regular supply of raw materials and other components. The regular flow of materials and supplies are helpful in the uninterrupted production.

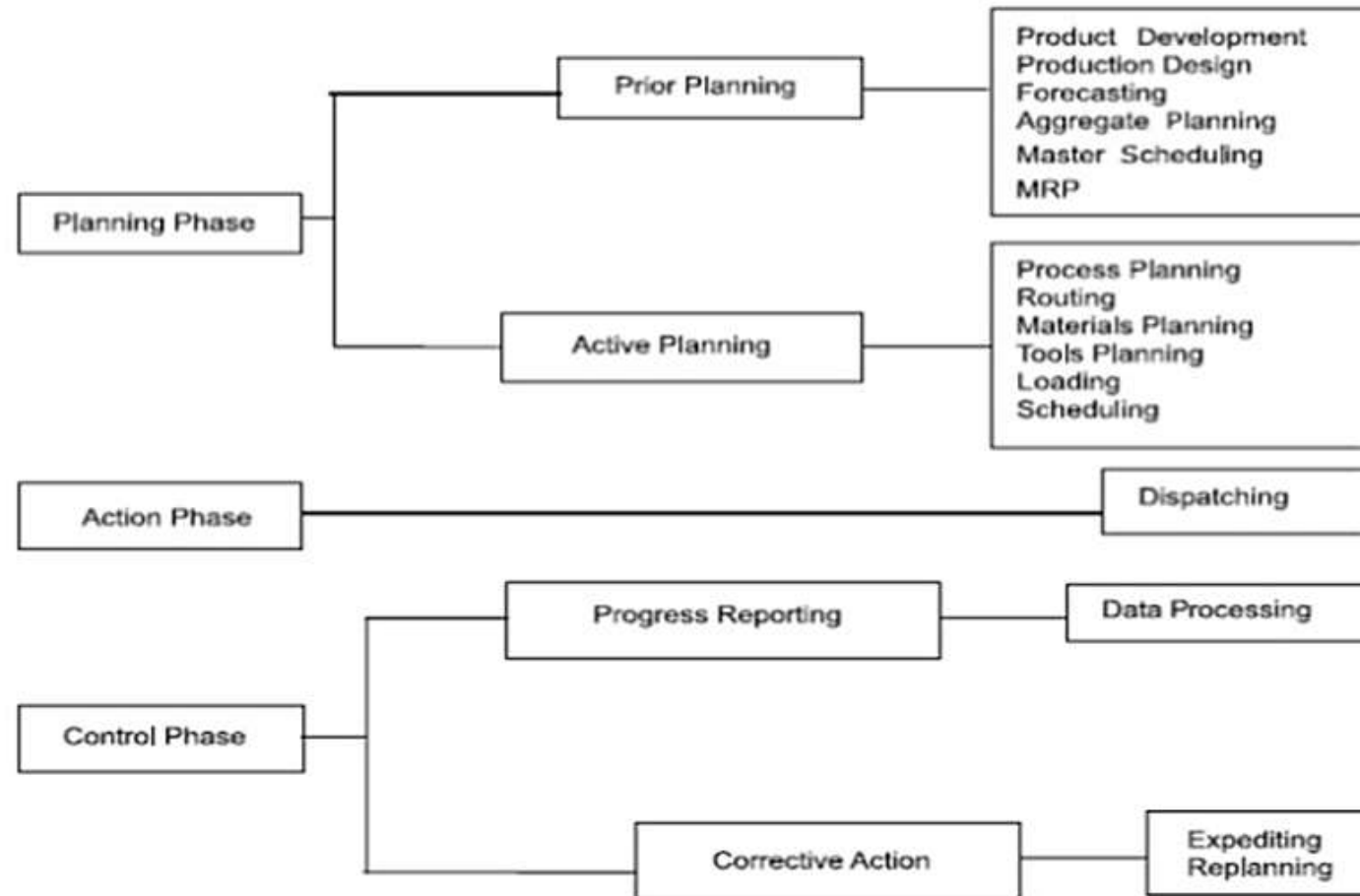
6. Capacity Utilization

There is a need to use the available resources effectively. It is helpful in bringing down various costs of production.

7. Timely Delivery

If there is good production planning and control, there will be timely production and the finished product will be rushed to the market in time. This also ensures the better relationship with the customers.

PRODUCTION PLANNING AND CONTROL HAS THREE PHASES NAMELY



Stages/Phases of production planning & control

❖ There are 3 stages or phases of production planning and control

1. Planning phase

A) **Pre planning phase** :- It involves product planning and development, facilities planning, plant planning, and plant location, etc.

B) **Active planning phase**:- It involves planning for quantity, determination of product mix, scheduling, materials, capacity planning, etc.

Stages continues...

2. Action Phase

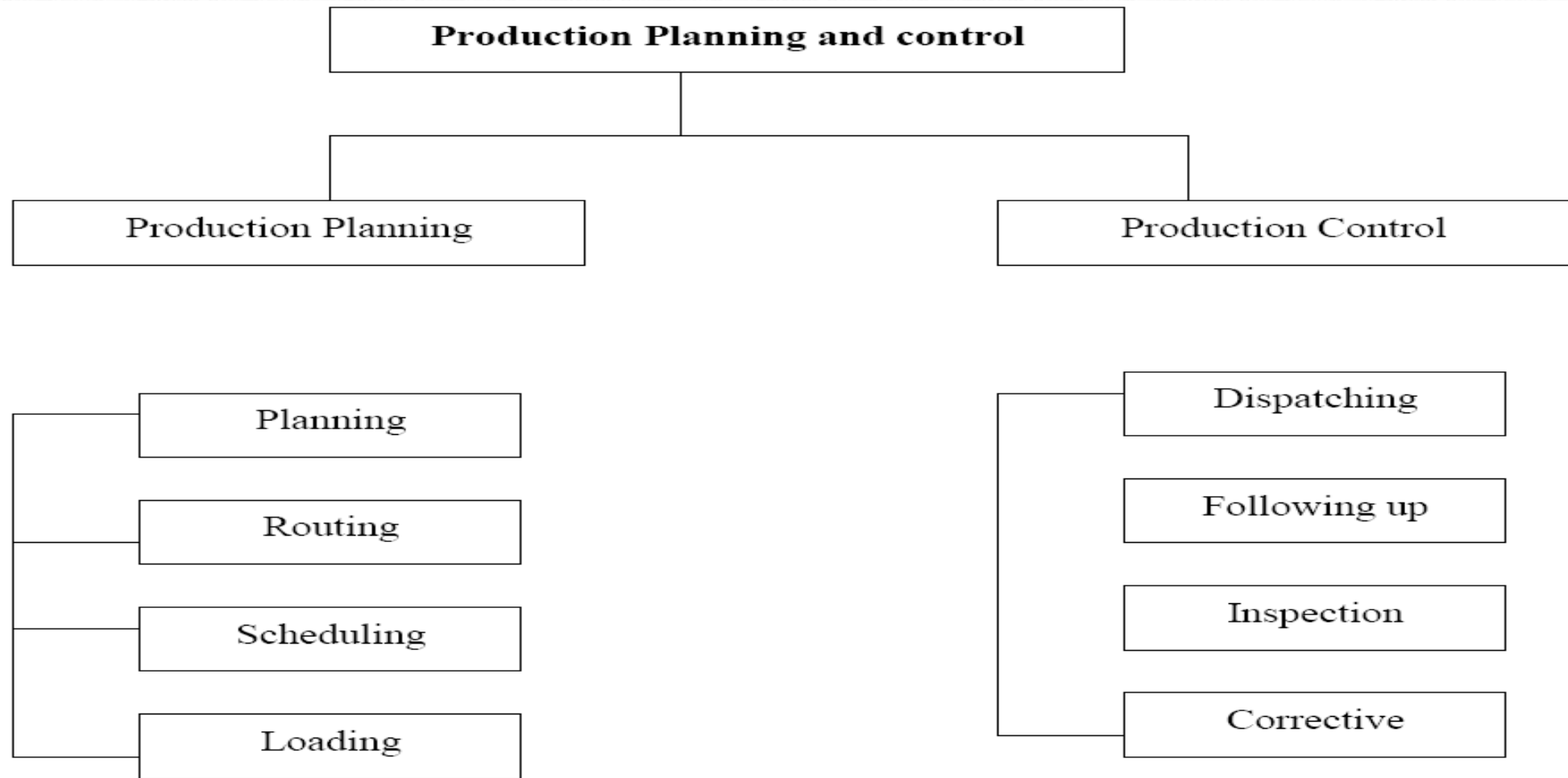
Action phase has the major step of dispatching. Dispatching is the transition from planning phase to action phase. Here the worker is ordered to start manufacturing the product. The tasks included in dispatching are job order, tool order, inspection order, etc.

Stages continues...

3. Control Phase

- A) **Progress Reporting :-** In progress reporting, the data regarding what is happening with the job is collected. It also helps to make comparison with present level of performance.
- B) **Corrective Action :-** The tasks under corrective action primarily make provisions for an unexpected event. For Example; make or buy decision, Schedule modifications, etc.

Functions of Production planning and control



Production planning

It is the planning of production and manufacturing modules in a company or industry. It utilizes the resource allocation of activities of employees, materials and production capacity, in order to serve different customers.

Production planning functions

1. Estimating

Involves deciding the quantity of products to be produced and cost involved in it on the basis of sales forecast. Estimating man power, machine capacity and materials required to meet the planned production targets are key activities before budgeting for resources.

2. Routing

Routing means, determination of route on which manufacturing operations will travel, establishing the sequence of operations to be followed in manufacturing a product.

Functions Continues...

3. Scheduling

It involves fixing priorities for each job and determining the starting time and finishing time for each operation, the starting dates and finishing dates for each part, Sub-assembly and final assembly. For Example; Total time required for the manufacture of a product, time table for production, etc.

4. Loading

Facility loading means loading of work center and deciding which jobs to be assigned to which work center or machine.

Production Control

Production control is the activity of monitoring and controlling any particular production or operation. Production control is often run from a specific control room or operations room.

Production control functions

1. Dispatching

The actual production starts with dispatching. It is a part of production control that translates the paper work into actual production in accordance with the details worked out under routing and scheduling functions.

2. Follow up

Follow up ensures that, the work is carried out as per the plan and delivery schedules are met. For example; Status reporting, controlling variations, monitoring progress of work through all stages of production, etc.

Control functions continues...

3. Inspection

Inspection is the process of examining an object for identification or checking it for verification of quality and quantity in any of its characteristics. It is an important tool for ascertaining and controlling the quality of a product.

Inspection can occur before production, during production, and after production.

Factors affecting production planning and control

- Use of computers
- Seasonal variations
- Variation in fashion and style
- Test marketing
- Vertical Vs Horizontal integration
- After sales service
- Losses due to predictable and unpredictable factors
- Production of order
- Design changes
- Rejection and replacement

Importance of production planning and control

- Higher productivity
- Removal of hurdles
- Better quality
- Consumer satisfaction
- saving in cost
- increase in production
- optimum utilization of capabilities
- Minimum over time
- Better industrial relations
- Better profitability

Limitations of production planning and control

1. Assumptions

It is based on certain assumptions or forecasts of consumer demand, plant capacity, availability of materials, tec.

2. Rigidity of employees behavior

Employees may resist changes in production levels set as per production plans if such plans are rigid.

3. Costly process

The implementation of production planning and control is costly

Limitations continues...

4. Time consuming process

The production planning process is time consuming when it is necessary to carry out routing and scheduling functions for large and complex products consisting of a large number of parts going into the product.

5. External environmental factors

It become extremely difficult when the environmental factors change very rapidly such as technology, customers taste and preferences, government policies, etc.

Master Production Schedule (MPS)

➤ Meaning of MPS

A master production schedule is a statement of what will be made, how many units will be made, and when they will be made. It is a production plan, not a sales plan. The MPS considers total demand on a plant's resources, including finished product sale, spare parts needs, and interplant needs. The MPS must also consider the capacity of the plant and the requirements imposed on vendors. Provisions are made in the overall plan for each manufacturing facility's operation. All planning for materials, manpower, plant, equipment, and financing for the facility is driven by the master production schedule.

Objectives of MPS

1. To provide top management with a means to authorize, and control manpower levels, inventory investment, and cash flow.
2. To coordinate marketing, manufacturing, engineering, and finance activities by a common performance objective.
3. To reconcile marketing and manufacturing needs.
4. To provide an overall measure of performance.
5. To provide data for material and capacity planning.

Functions of MPS

1. Translating aggregate plans

The aggregate plans sets the level of operations that roughly balances market demands with the material, labor and equipment capabilities of the firm. The aggregate is translated into specific number of end products to be produced in specific time period.

2. Evaluating alternative master schedules

Master scheduling is done on a trial and error basis. Trial fitting of alternative MPS can be done by simulation using computers. Detailed material and capacity required are then derived from the firm's MPS.

3. Effectively utilizing the capacity

By specifying the end item requirements over a time period, the MPS establishes the load and utilization parameters for labor and equipment.

Functions continues...

4. Generating material requirements

The MPS is the prime input to the MRP-1 system. The MRP-1 system provides for purchasing or manufacturing the necessary items in sufficient time to meet the final assembly dates specified, based on the MPS for end products.

5. Generating capacity requirements

Capacity needs, arise for manufacturing the components in the required time schedule to meet the requirements of end products as per the MPS. Capacity requirement planning is based on the MPS which should reflect an economic products as per the MPS. Capacity requirement planning is based on the MPS

Functions continues...

6. Facilitating information processing

By controlling the workload on work centres, the MPS determines the delivery schedules for end products for make-to-stock and make to order items. It coordinates other management information such as marketing capabilities, financial resources and personnel policies.

7. Maintaining valid priorities

The absolute or relative priorities for various jobs to be completed should reflect the true needs. This means that, the due date and the ranking of jobs should correspond with the time the order is actually needed. when customers change their orders or materials get scrapped sometimes, either the components are not actually needed or the end items cannot be produced because of shortage of some materials and then it is necessary that the MPS should be modified to reflect this change.



Menu X Sales Forecasts: SF/130 X Production Plan: PP/130 X Production Schedule: PS/130 X

Plan Reference : PP/1306/0004

Sale Forecast : SF/1305/0002

Period of Month : Juni

Start Date : 2013/06/01

End Date : 2013/06/30

Production Plan Schedule Notes

Description	UoM	On Hand	End of Month	Min	Max	Plan	Order	End Stock	Forecast n+1	Plan n+1	End Stock n+1	Forecast
[EA0162A05] AIR CUT VALVE ASSY	PCE	9.406,00	10.548,00	8.000,00	8.000,00	19.950,00	22.005,00	8.493,00	21.700,00	22.050,00	8.843,00	11.70
[EL0661A00] REED VALVE ASSY	PCE	3.094,00	5.195,00	12.000,00	12.000,00	31.500,00	23.205,00	13.490,00	21.700,00	21.000,00	12.790,00	11.70
[EA0146A00] AIR INDUCTION SYSTEM ASSY	PCE	1.227,00	1.277,00	1.000,00	1.000,00	500,00	600,00	1.177,00	806,00	1.000,00	1.371,00	70
[EA0170A02] AIR CUT VALVE ASSY	PCE	58,00	598,00	500,00	500,00	2.000,00	1.200,00	1.398,00	0,00	0,00	1.398,00	
[EA0205A02] AIR INDUCTION SYSTEM ASSY	PCE	69,00	69,00	1.000,00	1.000,00	2.200,00	800,00	1.469,00	1.700,00	1.650,00	1.419,00	60
[EL0492A00] VALVE ASSY 2ND AIR REED	PCE	5.572,00	6.508,00	8.000,00	8.000,00	26.600,00	24.420,00	8.688,00	29.940,00	29.400,00	8.148,00	24.12
[EA0227A00] VALVE ASSY AIR CUT	PCE	910,00	550,00	2.000,00	2.000,00	7.000,00	5.280,00	2.270,00	3.240,00	3.000,00	2.030,00	2.76
[EL0681A00] REED VALVE SUB ASSY	PCE	2.992,00	2.632,00	2.000,00	2.000,00	6.300,00	5.280,00	3.652,00	3.240,00	2.100,00	2.512,00	2.76
[EA0191A02] VALVE ASSY 2ND AIR	PCE	1.083,00	1.083,00	2.000,00	2.000,00	4.200,00	2.820,00	2.463,00	2.160,00	2.100,00	2.403,00	1.92
[EA0208A01] VALVE ASSY AIR CUT	PCE	6.455,00	3.600,00	5.000,00	5.000,00	18.900,00	16.980,00	5.520,00	1.800,00	1.800,00	5.520,00	1.92
[EA0250A01] VALVE ASSY AIR CUT	PCE	565,00	745,00	5.000,00	5.000,00	8.800,00	4.320,00	5.225,00	27.420,00	27.200,00	5.005,00	21.12
[EA0193A00] VALVE ASSY AIR CUT	PCE	1.027,00	1.027,00	1.000,00	1.000,00	1.000,00	120,00	1.907,00	720,00	0,00	1.187,00	1.08

Plan State : Approve

Validate

Create Schedule

No record selected

State:

THANK YOU

